MIL-S-1436
7 October 1949

MILITARY SPECIFICATION

SHELL, HE, M49 FOR 60 MM MORTAR
SHELL, PRACTICE, M50, FOR 60 MM MORTAR
SHELL, SMOKE, M302, FOR 60 MM MORTAR
SHELL, HE, M43, FOR 81 MM MORTAR
SHELL, PRACTICE, M43, FOR 81 MM MORTAR
SHELL, HE, M56, FOR 81 MM MORTAR
SHELL, CHEMICAL, M57, FOR 81 MM MORTAR
ASSEMBLING, MARKING AND PACKING

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-21-36
26 April 1946

This specification consists of this cover sheet and Specification 50-21-36, dated 26 April 1946, together with Amendment 1, dated 2 October 1946, attached hereto, without modification.

Copies of this specification may be obtained upon application to the Office, Chief of Ordnance, Washington 25, D. C.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

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U. S. ARMY
SPECIFICATION

No. 50-21-36
26 April 1946
Superseding
No. 50-21-11A dated 5/1/41
No. 50-21-26 dated 5-3-40
No. 50-21-29 dated 9-23-40

SHELL, HE, M49 FOR 60 MM MORTAR
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SHELL, CHEMICAL, M57, FOR 81 MM MORTAR
ASSEMBLING, MARKING AND PACKING

A. APPLICABLE SPECIFICATIONS.

A-1. The Ordnance Department drawings listed below pertaining to the round under contract, and all drawings and specifications listed or referred to thereon, form a part of this specification:

75-1-82 - SHELL, HE, M49A2, w/fuze, PD, M52A1, for 60 mm mortars
75-1-83 - SHELL, TP, M50A2, w/fuze, PD, M52A1, for 60 mm mortars
75-1-218- SHELL, Smoke, WP, M302, w/fuze, PD, M82, for 60 mm mortars
75-1-88 - SHELL, HE, M43A1, w/fuze, PD, M52A1, for 81 mm mortars
75-1-89 - SHELL, TP, M43A1, w/fuze, PD, M52A1, for 81 mm mortars
75-1-197- SHELL, HE, M56, w/fuze, TSQ, M77, for 81 mm mortars
75-1-97 - SHELL, HE, M56, w/fuze, PD, M53A1, for 81 mm mortars
75-1-94 - SHELL, SMOKE, FS, M57, w/fuze, PD, M52A1, for 81 mm mortars
75-1-198- SHELL, SMOKE, FS, M57, w/fuze, TSQ, M77, for 81 mm mortars
75-1-93 - SHELL, SMOKE, WP, M57, w/fuze, PD, M52A1, for 81 mm mortars
75-1-199- SHELL, SMOKE, WP, M57, w/fuze, TSQ, M77, for 81 mm mortars.

A-2. All drawings and specifications shall be of the issue in effect on the date of invitation for bids.

B. TYPES AND CLASSES.

B-1. This specification covers 3 types as follows:
Type A - HE Loaded Shell.
Type B - Inert Loaded Shell with Black Powder Pellet (Practice).
Type C - Chemical Loaded Shell.

C. MATERIAL AND WORKMANSHIP.

C-1. Materials and parts shall be in accordance with applicable specifications and drawings.

C-2. All parts shall be free from chips, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

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C-3. The painting and markings shall be restored by the assembling contractor on any components furnished to him painted and marked, when such painting and markings have been marred or obliterated.

C-4. Visual inspection of components. All components and materials furnished by the United States shall be given a careful visual inspection by the contractor before assembling, to prevent the possible use of any damaged or obviously defective parts. If any such parts are found, they shall be submitted to the inspector for his examination and decision as to their fitness for service.

C-5. The method of loading and assembling shall be uniform for each lot.

C-6. Cleaning of shell threads (type A and type B shell). After loading of the shell has been completed, the threads of the shell shall be thoroughly cleaned of explosives.

C-6a. Type A shell. No liquid other than acetone shall be used as a cleaner. If acetone is used, there shall be no contamination of the shell charge.

C-6b. Type B shell. No liquid shall be used as a cleaner.

D. GENERAL REQUIREMENTS.

D-1. See section E.

E. DETAIL REQUIREMENTS.

E-1. The rounds shall meet all requirements prescribed on drawings and in applicable specifications.

E-2. Proving ground. The round shall meet the following requirements in the proving ground tests:

E-2a. Malfunctioning of components. There shall be no malfunctioning of components above that permitted by the respective detail specifications for such components.

E-2b. In the proving ground test a score of not less than 3600 shall be obtained in accordance with each of tables I, II, and III for every 40 rounds fired. (See paragraph F-4d.)

F. METHODS OF SAMPLING, INSPECTION AND TEST.

F-1. Inspection.

F-1a. Inspection shall be as specified in U. S. Army General Specification for Ammunition Except Small Arms Ammunition, No. 50-0-1.

F-1b. Gaging. After the round has been assembled, the contractor shall gage each round of ammunition in a profile and alignment gage (commonly known as a chamber gage), pertaining to this round. If the round fails to enter the profile and alignment gage freely, it shall be rejected subject to correction and retest. Drawings of the Government profile and alignment gage will be furnished to the contractor for his information upon application to the inspector.

F-2. Round lots.

F-2a. Lot sizes.

F-2a(1). Maximum, 20,000.

F-2a(2). Minimum, 2,000.

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F-2b. Lot content. Each lot shall contain:

Loaded shell limited to one manufacturer, same drawing, same drawing revision, and same specification and specification revision.

Loaded powder train type time fuzes from one lot, or

Loaded point detonating fuzes from not more than 2 lots or 2 grand lots (if grand lot assignments have been made for the item).

Fin assemblies made by one manufacturer at one plant in accordance with the same drawing.

Loaded ignition cartridges from not more than 2 lots or 2 grand lots (if grand lot assignments have been made for the item). These lots shall be consecutive lots, if possible (not a requirement).

Loaded primers from not more than 2 lots or 2 grand lots (if grand lot assignments have been made for the item). These lots shall be consecutive lots, if possible (not a requirement).

For type C shell, burster charges from not more than 2 lots or 2 grand lots (if grand lot assignments have been made for the item). These lots shall be consecutive lots, if possible (not a requirement).

Propellent increments from not more than 2 lots. These shall be consecutive lots, if possible.

Rounds of the same class.

F-3. Sampling. Forty rounds shall be selected by the inspector from each lot for proving ground tests.

F-4. Proving ground tests. The tests shall be conducted as follows:

F-4a. Zones.

F-4a(1). Low. Fire 20 of the sample rounds, in a mortar for which the round is standard in the lowest zone which is standard for the round.

F-4a(1)a. M49, M50 and M43 complete rounds. Ignition cartridge only.

F-4a(1)b. M56 and M57 complete rounds. Ignition cartridge and one increment.

F-4a(2). High. Fire 20 of the sample rounds, in a mortar for which the round is standard, with ignition cartridge and 4 increments.

F-4b. Velocity. Fire the first 5 rounds in each zone, (see par. F-4a) through solenoid coils and measure the velocity. For each zone firing, record the computed average velocity and the maximum velocity dispersion obtained. This information shall be recorded on the firing record for information purposes only.

F-4c. Time fuzes. When testing rounds assembled with Fuze, TSQ, M77, fire 10 rounds in each zone with fuzes set and the mortar elevated to correspond to one of the conditions for acceptance firing of the fuze as given in the fuze specification. For each zone firing, measure the time to point of burst, and record the computed mean time and maximum time dispersion. No specific requirements for mean time and dispersion will be required, but the results will be compared with the requirements as stated in the fuze specification; failure to obtain reasonable correspondence shall cause the lot to be rejected and referred to the Office, Chief of Ordnance, for disposition.

F-4d. Scoring. Score the results of the proving ground test, and make the necessary observations and measurements in accordance with tables I, II, and III. Whenever two or more types of failures occur in the same round, the major penalty only shall be deducted.

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Method of Scoring

Each round is given a score of 100 minus such deductions as are listed in each of tables I, II, and II.

TABLE I

		Deduct
1.	Rounds functions satisfactorily	0
2.	Horizontal range less than 80 percent, but more than 60 percent, of the expected range	200
3.	Fin blade became detached but horizontal range is 80 percent, or more, of the expected range	100
4.	Shell fails to burst	200
5.	Fuze set for air burst but functions on ground impact	100
6.	Shell (Type A) functions low order	150
7.	Shell (Type B or C) fails to produce a smoke cloud	150
8.	Premature burst (see par. H-3)	4000

TABLE II

1.	Horizontal range less than 60 percent of the expected range	800

TABLE III

1.	Misfire	800

F-4e. Retest. A retest using double the number of samples used in the original test may be made, if requested by the contractor, provided the total score in any or all the tables I, II, and III is less than 3600 but not less than 3200. The scores for the original test plus the retest, for each of tables I, II, and III, must be 10,800 or greater to permit acceptance.

G. PACKAGING, PACKING, AND MARKING.

G-1. Packaging. As specified on the drawing.

G-2. Packing.

G-2a. As specified on the drawing.

G-2b. Data cards. Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted rounds in accordance with the form specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

G-2b(1). Form to be used for data cards. (Note: Headings need be made only for applicable components.)

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AMMUNITION DATA CARD

Kind_____ Ammunition Lot No._____
AIC Symbol_____ Quantity in Lot_____
Round Drg. No.*_____ Spec. No.*_____
Assembled by_____ Date_____
Weight of projectile as fired_____
Contract No._____ Order No._____

Expected Muzzle Velocities and Pressures. (Proof weight of projectile on which the ballistic data are based_____pounds.)

Zone	Wt. Chg. by Zones (Grains)	Total Wt. Chg. Grains	Muzzle Velocity f.s.	Pressure Lbs. per Sq. Inch
1.				
2.				
3.	(Provision shall be made for 4 zone charges for M56 and M57 Complete Rounds, 5 zone charges for M49 and M50 Complete Rounds and 7 zone charges for M43 Complete Rounds.)			
4.				
5.				
6.				
7.				

Packed_____in a box.
Remarks_____

COMPONENTS

	Kind	Manufacturer	Lot No.	Year Mfgd	Date Filled	Method	Drg. No.*
Primer							
Ignition Cartridge							
Shell							
Fin Assembly							
Filler							
Burster Casing							
Black Powder Pellet[1]							
Prop. Incr.							
Fuze							
Fiber Container							
Metal Container							

Certified to by_____, Inspector
(Print here name of inspector, district, arsenal or loading plant, and sub-district, where applicable.)

[1] For Type A complete rounds - omit.
For Type C complete rounds - change to read "Burster Charge".

* Include revision, amendments, and ANR's.

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G-2c. Firing table card. A firing table card, properly filled out with the data as shown on the Abridged Firing Table applicable to the complete round being packed, shall be placed in each box. The card shall be of the same size as the Ammunition Data Card and shall present data in substantially the same form as shown on the applicable Abridged Firing Table.

G-3. Marking. The marking for shipment shall be as specified on the drawing.

H. NOTES.

H-1. Round. A round as referred to herein means the assembly of all the components of ammunition into one unit necessary for firing the round from a mortar; example - loaded ignition cartridge, primer, propellent increments, fin assembly and loaded shell and loaded fuze, all properly assembled.

H-2. When the contractor for the round is also the contractor for one or more of the components thereof, the proving ground tests of the components may be combined with the proving ground tests of the round to save expense, upon agreement between the United States and the contractor to that effect. In case the round specification does not cover all of the proving ground tests specified for the component the additional tests prescribed in the component specification will be conducted.

H-3. Premature bursts.

H-3a. Rounds with M52 and M53 PD fuze. A "premature burst" is defined as a functioning of the shell in the mortar bore or in flight before impact.

H-3b. Rounds with M77 TSQ fuze. A "premature burst" is considered to have occurred if the time of flight is less than 50 percent of the mean time of flight as determined from the other rounds tested at the same fuze setting.

H-4. This specification also replaces Ordnance Department Tentative Specification AXS-1466.

H-5. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE: This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

NOTE: Copies of this specification may be obtained from the Office, Chief of Ordnance, Army Service Forces, Washington 25, D. C.

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U. S. ARMY
SPECIFICATION

No. 50-21-36
26 April 1946
Amendment 1
2 October 1946

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ASSEMBLING, MARKING AND PACKING

The following changes shall be made in Specification No. 50-21-36, dated 26 April 1946:

Page 4, Table I. Delete item 2 and substitute the following:
"2. Horizontal range less than 80 percent, but more than 60 percent, of the average range - - - - - - - - - - - - 200"

Page 4, Table II. Delete item 1 and substitute the following:
"1. Horizontal range less than 60 percent of the average range - - -800"

Page 6, Section H. Add a new paragraph as follows:
"H-6. In determining short rounds for purposes of scoring in Tables I and II, the average range will be compouted after excluding rounds whose horizontal range is less than 80 percent of this average. Should this average range deviate from the expected range by more than 10 percent, it will be reported to the Chief of Ordnance for special consideration."

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